Anna M. Maddux

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Education

École Polytechnique Fédérale de Lausanne (EPFL) Feb 2022 – Present
PhD in Multi-agent learning
- PhD supervisor: Maryam Kamgarpour

Eidgenössische Technische Hochschule Zürich (ETHZ) Sept 2019 – May 2021
M.S. in Statistics
- Master's Thesis: "Behaviour Estimation in Dynamic Games" (https://www.researchcollection.ethz.ch/handle/20.500.11850/485983)
- Courses: Fundamentals of Mathematical Statistics, Optimization for Data Science, Advanced Machine Learning, Computational Statistics

Albert Ludwig University of Freiburg Oct 2015 – March 2019
B.S. in Mathematics
- Bachelor's Thesis: "Symmetric Random Walks and a Gerneralization of the Borel-Cantelli Lemma"

Albert Ludwig University of Freiburg Oct 2013 – Feb 2019
B.S. in Economics
- Bachelor's Thesis: "Hyperparameter Optimization of Neural Networks: Grid Search versus Random Search"

Research and Work Experience

ETH Zürich | *Research Assistant* Oct 2021 – Dec 2021
Computational Social Science (COSS) Group under the Supervision of Dirk Helbing

ETH Zürich | *Research Assistant* May 2021 – July 2021
Automatic Control Laboratory (IfA) Group under the Supervision of Florian Dörfler

Daimler Benz Stuttgart | *Data Science Intern* April 2019 – Aug 2019
Advanced Analytics and Big Data Group

German Bundestag Berlin | *Research Intern* Jan 2019 – Feb 2019
Supervisor: Dr. Christoph Hoffmann, Member of the German Parliament

Fellowships

Accenture Female Talent Program March 2020 – Oct 2020
Mentoring program to promote females in STEM and economics

Fellow of the Friedrich Naumann Foundation for Freedom Oct 2017 – May 2021
Academic scholarship throughout my undergraduate and graduate studies (held by 1% of German university students)

Publications

* authors contributed equally; + mentored student

Conference and Journal Publications

Anna M Maddux, Maryam Kamgarpour. Multi-Agent Learning in Contextual Games under Unknown Constraints. Artificial Intelligence and Statistics Conference (AISTATS), 2024.

Gabriel Vallat+, Jingwen Wang, **Anna M Maddux**, Maryam Kamgarpour, Stefana Parascho. Reinforcement learning for scaffold-free construction of spanning structures. ACM Symposium on Computational Fabrication (ACM SCF), 2023.

Anna M Maddux, Nicolò Pagan, Giuseppe Belgioioso, Florian Dörfler. Data-driven behaviour estimation in parametric games. International Federation of Automatic Control (IFAC), 2023.

Preprints and under Submission

Anna M Maddux*, Reda Ouhamma*, Maryam Kamgarpour. Finite-time convergence to an ϵ-efficient Nash equilibrium in potential games. In submission (arXiv:2405.15497), 2024.

Professional Activities, Leadership, and Service

Conference and Workshop Organization

Workshop Organizer, Workshop on Entrepreneurship and Innovation | *Beatenberg, Switzerland* April 2024
Junior retreat of the NCCR Automation

Peer Review

International Conference on Machine Learning (ICML) Workshop,International Federation of Automatic Control (IFAC), European Control Conference (ECC), Conference on Decision and Control (CDC)

Teaching

University Courses

Teaching Assistant, Multi-Agent Learning and Control | *EPFL Lausanne* Spring 2024
Teaching Assistant, Foundations of Artificial Intelligence | *EPFL Lausanne* Fall 2023
Teaching Assistant, Foundations of Artificial Intelligence | *EPFL Lausanne* Fall 2022
Teaching Assistant, Analysis 2 | *ETH Zürich* Spring 2020
Teaching Assistant, Microeconomics 1 | *Albert Ludwigs University of Freiburg* Fall 2016
Teaching Assistant, Microeconomics 2 | *Albert Ludwigs University of Freiburg* Spring 2016
Teaching Assistant, Microeconomics 1 | *Albert Ludwigs University of Freiburg* Fall 2015

Science Education

Presentation at the Kantonsschule Beromünmster | *Beromünster, Switzerland* Feb 2024
Presentation under the NCCR Automation initiative "A researcher in my class".

Mentoring

Antoine Bergerault, Semester Project | *EPFL Lausanne* Feb 2023 - June 2024
Title: "Learning to maximize the social welfare from preference feedback".
Co-supervision with Andreas Schlaginhaufen (EPFL), and Maryam Kamgarpour (EPFL).

César Camus-Emschwiller, Semester Project | *EPFL Lausanne* Feb 2023 - June 2024
Title: "Learning in potential games with coupling constraints - Practical application".
Co-supervision with Gabriel Vallat (EPFL), and Maryam Kamgarpour (EPFL).

Julien Drouet, Semester Project | *EPFL Lausanne* Feb 2023 - June 2024
Title: "Learning in Potential Games with Coupling Constraints".
Co-supervision with Gabriel Vallat (EPFL), and Maryam Kamgarpour (EPFL).

Sabri El Amrani, Semester Project | *EPFL Lausanne* Oct 2023 - Jan 2024
Title: "Reinforcement Learning from Human Feedback for the Construction of Spanning Structures".
Co-supervision with Andreas Schlaginhaufen (EPFL), and Maryam Kamgarpour (EPFL).

Dana Afazeli, E3 Summer Internship | *EPFL Lausanne* Jun 2023 - Sept 2023
Co-supervision with Kai Ren (EPFL), and Maryam Kamgarpour (EPFL).

Titouan Renard, Master Project | *EPFL Lausanne* Feb 2023 - July 2023
Title: "Provable Convergence Guarantees for Constrained Inverse Reinforcement Learning".
Co-supervision with Andreas Schlaginhaufen (EPFL), Tingting Ni (EPFL), and Maryam Kamgarpour (EPFL).

Riccardo Ruzza, Bachelor Project | *EPFL Lausanne* Feb 2023 - June 2023
Co-supervision with Giulio Salizzoni (EPFL), and Maryam Kamgarpour (EPFL).

Julien Gouraud, Bachelor Project | *EPFL Lausanne* Feb 2023 - June 2023
Co-supervision with Giulio Salizzoni (EPFL), and Maryam Kamgarpour (EPFL).

Gabriel Vallat, Master Project | *EPFL Lausanne* Feb 2023 - July 2023
Title: "Reinforcement learning for scaffold-free construction of spanning structures".
Co-supervision with Maryam Kamgarpour (EPFL), and Stefana Parascho (EPFL). Resulting paper accepted at ACM 2023.

Dorsa Majdi, E3 Summer Internship | *EPFL Lausanne* Feb 2023 - July 2023
Co-supervision with Maryam Kamgarpour (EPFL).

Languages

Computer Languages: Python, Matlab, LaTeX
Languages: German (mother tongue), English (fluent), French (B1)